

Initial Project System Representation

I. Project Information

Project Title: Passenger Queue Dynamics and Waiting-Time Simulation for Selected M66 Bus Stops

Group Members

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II. Problem Statement & Objective

Passenger demand and bus arrival intervals change during the day, so queue length and waiting time may differ from one M66 stop to another.

The simulation aims to estimate passenger queue length and waiting time at five consecutive M66 eastbound stops during weekday morning (6:00–11:00 AM) and evening (3:00–8:00 PM) periods. The project will use schedule and ridership data from the Metropolitan Transportation Authority (MTA) for the M66 bus route, including General Transit Feed Specification (GTFS) data. Using these data and a 60-passenger bus limit, the simulation will report minimum, average, and maximum waiting time per stop and compare both periods. An AI model will predict hourly boardings for each stop.

III. System Representation

Element	Description
System	Queueing-and-bus-loading system
Agents	Passengers and M66 buses.
Environment	W 65 St/Central Park West, E 65 St/5 Ave, Madison Ave/E 65 St, Madison Ave/E 67 St, E 68 St/Lexington Ave, and the route segment connecting these stops during the selected weekday periods.

State Variables	queue_length, queue_entry_times, next_bus_arrival, hourly_demand (observed or set by a rounded AI prediction), bus_occupancy, and available_space. Fixed parameter: bus_capacity = 60 passengers.
Events	Passenger arrival, bus arrival, passenger alighting, passenger boarding, and queue update.
Outputs	Queue length over time in passengers, minimum, average, and maximum estimated waiting time in minutes, and morning-evening comparison.

IV. State Definition

The state represents the current condition of the passenger queues and simulated buses at the five selected stops at any given time. The fixed capacity parameter is also included because it limits passenger boarding.

State/Parameter	Description	Data Type	Initial Value
queue_length	Passengers waiting at each selected stop	Integer	0 at each stop
queue_entry_times	Times when waiting passengers joined each queue	List	Empty list at each stop
next_bus_arrival	Next scheduled GTFS bus arrival at each stop	Timestamp	First scheduled arrival in the period
hourly_demand	Observed or AI-predicted boardings for each stop and hour	Integer	Observed count or rounded AI prediction
bus_capacity	Maximum passengers allowed inside each simulated bus	Integer	60 passengers
bus_occupancy	Passengers currently inside each simulated bus	Integer	Estimated before the first selected stop
available_space	Passengers who can still board: capacity minus occupancy	Integer	60 minus bus occupancy

Separate values and lists are maintained for the five selected bus stops. Bus capacity is fixed, while queue length, demand, bus occupancy, and available space can change.

A timestamp stores the date and scheduled time of an event. Passenger quantities are whole-number counts and cannot be negative. The AI may produce a decimal prediction, so it is rounded to the nearest whole passenger and any negative value is changed to 0 before it becomes hourly_demand and virtual passengers are generated.

Estimated waiting time is calculated when a passenger boards: boarding time minus queue entry time. It is stored as an output record rather than treated as a current state variable. A bus boards only the number of passengers that fit in its available space. Other passengers remain in the queue and continue waiting.

Capacity and occupancy basis

Every simulated bus is treated as a standard MTA bus with a hard limit of 60 passengers. An official MTA glossary states that a standard bus has a 60-passenger capacity. Because the dataset does not identify the exact vehicle assigned to every M66 trip, this value is used as a clear modeling assumption; actual capacity may vary by vehicle.

The stop-level MTA dataset contains boardings and alightings. Occupancy before the first selected stop is estimated by dividing the cumulative net passengers from earlier eastbound M66 stops by the recorded trips for that hour. The result is rounded and kept between 0 and 60. At each selected stop, hourly alightings are divided among the buses recorded for that hour, rounded to whole passengers, and limited so they cannot be greater than the bus's current occupancy.

Data used by the model

The stop-level ridership data provide the date, hour, route, direction, stop, stop sequence, boardings, alightings, and recorded trips. MTA GTFS provides stop names, stop order, scheduled trips, and scheduled bus-arrival times.

Capacity Reference: Metropolitan Transportation Authority, Second Avenue Subway FEIS Glossary, p. Glossary-2 (standard bus capacity of 60 passengers):
<https://www.mta.info/document/22336>

Ridership Data Reference: MTA Bus Stop Level Ridership: Beginning 2024 (boardings and alightings): <https://data.ny.gov/Transportation/MTA-Bus-Stop-Level-Ridership-Beginning-2024/fvdm-uavx>

Schedule Reference: MTA Developer Resources, GTFS schedule data: <https://www.mta.info/developers>

Interpretation and limits: The public data do not contain individual passenger-arrival times, measured queues, or actual waiting times. Historical boardings are therefore used as a practical estimate of passenger demand, although they may not include people who were unable to board. Virtual arrival times are generated within each hour. GTFS contains scheduled rather than actual bus arrivals. The results are simulation estimates, not claimed measurements of past passenger waits.

For the proposed AI component, the following variables will be used to predict hourly passenger boardings:

AI Variables	Description
stop_id	Input that identifies the selected M66 bus stop
hour_of_day	Input that identifies the simulated hour
travel_period	Input: morning or evening
day_of_week	Input: Monday through Friday
scheduled_trip_count	Input: scheduled M66 trips during the hour
past_average_boardings	Input: average boardings for the same stop and hour from earlier dates only
predicted_hourly_boardings	AI output; rounded to a whole-number hourly_demand value

V. Simulation Question

How do passenger queue lengths and estimated waiting times at five selected M66 eastbound bus stops change over time under varying passenger demand and scheduled bus-arrival intervals when each bus is limited to 60 passengers?

VI. Simulation Logic

Morning and evening are simulated separately. Each simulation step represents a passenger or bus event.

1. For each stop and hour, observed boardings or a rounded AI prediction provide the whole-number passenger-demand total. That number of passengers is spread randomly across the 60 minutes.
2. A passenger joins the queue at the selected stop and records the queue entry time.
3. Each eastbound M66 bus arrives according to its GTFS schedule. Its occupancy before the first selected stop is estimated from earlier M66 boardings and alightings.
4. Estimated alighting passengers leave the bus without exceeding its current occupancy.
5. Available space is calculated as 60 minus the current bus occupancy. Waiting passengers board only until the bus reaches 60 passengers.
6. Passengers who cannot board remain in the queue. Internal occupancy is updated, and the waiting time of each boarded passenger is recorded for the final results.

Each scheduled eastbound GTFS trip is treated as one bus. Return trips are outside the project boundary because westbound trips are separate GTFS records. After each selected period ends, no new passengers are generated, but scheduled buses continue until the remaining queue is cleared. Morning and evening runs are repeated and averaged so that one random set of virtual arrival times does not determine the conclusion.

VII. Preliminary AI Justification

Selected AI Role: Prediction

AI Component: A passenger-demand prediction model

AI Output: Predicted hourly passenger boardings for each selected stop. The decimal prediction is converted to a non-negative whole passenger count before simulation.

AI Inputs: The stop, hour, travel period, day, scheduled trip count, and past boarding average listed in the State Definition.

How it supports the simulation: The predicted hourly boarding total becomes the passenger-demand input used by the simulation.

Why prediction is useful: Passenger demand changes by stop, hour, and day. Predicting demand allows the simulation to represent changing conditions instead of using one fixed passenger count.

How it will be checked: Train the AI on earlier dates, then compare its predicted hourly boardings with actual MTA boarding counts from later dates that were not used for training. The closer the values are, the more accurate the prediction is.

AI Boundary: AI predicts passenger demand only. The bus capacity and scheduled arrivals come from MTA references and data, not from AI-generated values.

VIII. Expected Outputs

Simulation Outputs

- Queue length in passengers for each selected bus stop, including passengers left waiting after a bus departs
- Minimum, average, and maximum estimated waiting time in minutes for each selected bus stop
- Morning and evening comparison

AI Outputs

- Predicted hourly passenger boardings for each selected bus stop, converted to a non-negative whole passenger count

Visualizations

- Queue-length graph for each selected bus stop
- Waiting-time comparison graph
- Morning-versus-evening comparison graph

IX. Group Responsibilities

Member	Responsibility
Elijah Crisehea M. Nollen	System & State Lead
Ma. Sofia Anne C. Revilla	Logic & Simulation Lead
Karol Joy M. Ronquillo	Documentation & QA Lead

X. AI Use Guideline and Disclosure

AI Tool(s) Used

- Gemini

Purpose of Use

- Improving the clarity and completeness of the project description
- Correcting grammar and sentence structure
- Explaining technical concepts related to modeling, simulation, and AI

Description of Contribution

Gemini was used to refine the wording and organization of the project description, correct grammatical errors, and explain technical concepts such as state variables and AI-based demand prediction. All generated content was reviewed, edited, and validated by the group.

AI Responsibility Statement

The group remains responsible for:

- Selecting and properly using the datasets
- Defining the system design and project scope
- Establishing the simulation logic and assumptions
- Selecting, justifying, evaluating, and explaining the limitations of the AI model
- Ensuring the accuracy of the documentation and final submission

The use of AI tools does not transfer responsibility for the submitted work.

Ethical and Responsible AI Consideration

The project uses publicly available, aggregated MTA data and does not include names or other personal information. Virtual passenger-arrival times are generated only for the simulation and do not represent identified individuals. AI predictions will be clearly labeled as estimates and evaluated using actual aggregated MTA data.